

AMES, IA, USA

WMO: 725472

Lat: 41.991N

Lon: 93.619W

Elev: 955

StdP: 14.20

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 94989

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.7	(c) -1.6	(d) -13.4	(e) 2.8	(f) -5.8	(g) -9.0	(h) 3.5	(i) -0.6	(j) 30.9	(k) 21.6	(l) 27.3	(m) 19.8	(n) 7.9	(o) 310	(p) 0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.8	(c) 90.8	(d) 76.1	(e) 88.3	(f) 74.8	(g) 85.8	(h) 73.2	(i) 79.1	(j) 87.3	(k) 77.2	(l) 85.3	(m) 75.4	(n) 82.9	(o) 11.8	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.7	144.3	84.5	74.7	134.8	82.5	72.9	126.4	80.4	43.3	87.5	41.4	85.2	39.6	83.0	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.2	(b) 23.5	(c) 20.2	(e) -16.1	(f) 95.0	(g) 6.4	(h) 2.5	(i) -20.7	(j) 96.9	(k) -24.4	(l) 98.3	(m) -28.0	(n) 99.7	(o) -32.7	(p) 101.6		
(4) 26.2	(5) 23.5	(6) 20.2	(7) -16.4	(8) 82.3	(9) 6.3	(10) 1.8	(11) -20.9	(12) 83.6	(13) -24.6	(14) 84.6	(15) -28.1	(16) 85.6	(17) -32.6	(18) 86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.2	20.9	24.3	37.1	50.2	61.3	70.6	74.0	71.0	64.5	51.5	38.1	25.6	
	DBStd	20.93	12.18	12.47	11.84	9.47	8.26	6.09	5.39	5.19	7.67	9.48	10.30	11.65	
	HDD50	3403	903	720	429	110	10	0	0	0	2	97	375	757	
	HDD65	6586	1368	1139	867	451	171	19	3	9	102	430	806	1221	
	CDD50	3117	0	1	28	115	360	617	745	652	436	143	19	1	
	CDD65	826	0	0	1	6	56	186	283	197	86	11	0	0	
	CDH74	7860	0	0	12	117	556	1755	2704	1688	887	137	4	0	
	CDH80	2549	0	0	0	25	142	592	980	506	272	32	0	0	
Wind	WSAvg	9.2	10.2	10.4	10.8	11.4	10.2	8.5	6.7	6.1	7.4	9.0	10.0	9.9	
Precipitation	PrecAvg	35.8	0.9	1.0	2.2	3.9	5.2	5.2	4.8	4.4	2.9	2.3	1.9	1.1	
	PrecMax	55.8	1.8	1.9	6.3	8.0	9.1	12.7	13.6	13.4	5.4	7.6	4.7	2.4	
	PrecMin	23.2	0.4	0.2	0.1	1.1	1.4	0.8	1.2	0.7	0.7	0.2	0.1	0.0	
	PrecStd	7.9	0.4	0.5	1.5	1.6	2.0	3.1	3.0	3.1	1.4	1.8	1.2	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	61.4	75.3	83.8	89.5	92.5	94.5	92.5	90.5	83.9	72.6	58.9	
		MCWB	43.8	52.4	61.4	64.1	71.2	74.7	79.0	76.1	73.1	68.1	58.5	52.8	
	2%	DB	45.4	52.2	67.9	77.3	83.8	89.6	91.1	88.4	87.2	77.9	64.6	50.5	
		MCWB	39.3	46.2	56.6	61.4	67.7	73.9	78.2	76.5	72.3	63.4	55.0	45.6	
	5%	DB	40.0	45.7	61.4	72.3	80.5	86.4	88.4	85.1	82.5	72.6	59.0	45.6	
		MCWB	36.3	41.0	53.2	57.7	65.7	71.9	76.7	74.4	69.6	61.6	50.7	41.0	
	10%	DB	36.2	39.9	54.9	67.2	76.3	83.3	85.5	82.3	79.4	67.9	54.2	40.7	
		MCWB	33.7	36.6	47.7	55.2	63.6	70.4	74.8	72.6	67.3	58.5	47.8	37.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.8	53.7	63.3	67.0	74.1	79.2	82.4	80.5	76.9	71.2	60.5	54.2
			MCDB	52.3	59.8	71.6	79.3	85.4	89.2	90.8	87.9	86.6	78.2	67.7	57.3
2%		WB	40.2	46.8	59.1	63.2	71.0	76.7	80.1	77.7	74.4	67.1	56.9	47.1	
		MCDB	45.2	51.6	65.0	72.7	80.4	85.7	88.2	85.6	82.7	73.7	64.1	48.8	
5%		WB	36.7	41.1	53.8	60.0	68.3	74.6	78.2	76.0	72.4	63.7	52.3	41.4	
		MCDB	39.7	45.5	60.9	69.7	77.0	83.1	86.0	83.3	79.9	68.9	57.7	45.6	
10%		WB	33.8	37.0	48.2	56.7	66.0	72.6	76.3	74.2	69.8	60.1	47.9	37.3	
		MCDB	35.9	40.1	55.0	65.6	73.6	80.6	83.6	80.8	76.6	67.1	53.6	40.6	
Mean Daily Temperature Range		5% DB	MDBR	16.7	17.0	18.5	21.7	20.7	20.2	19.8	20.1	23.6	22.1	19.2	16.3
			MCDBR	22.7	22.8	27.9	30.5	25.3	23.5	21.7	22.9	26.8	28.6	28.0	22.5
	MCWBR		17.5	17.1	17.5	17.0	12.5	11.3	10.9	12.3	13.5	15.8	17.8	17.4	
	5% WB	MCDBR	20.9	22.3	25.7	25.3	21.6	20.3	19.7	20.2	22.2	22.0	25.1	20.9	
		MCWBR	16.6	17.1	16.9	15.9	12.2	11.9	11.1	11.9	12.5	14.7	17.6	16.9	
Clear-Sky Solar Irradiance	taub	0.287	0.308	0.327	0.384	0.413	0.425	0.441	0.423	0.391	0.344	0.301	0.287		
	taud	2.415	2.358	2.408	2.272	2.260	2.272	2.209	2.268	2.311	2.437	2.499	2.436		
	Ebn at Noon	274	284	292	281	274	270	264	266	266	268	268	264		
	Edh at Noon	25	31	33	41	43	42	44	41	36	28	23	22		
All-Sky Solar Radiation	RadAvg	602	922	1201	1504	1776	1974	2006	1728	1437	961	652	495		
	RadStd	61	66	116	116	123	136	177	134	115	103	117	53	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page